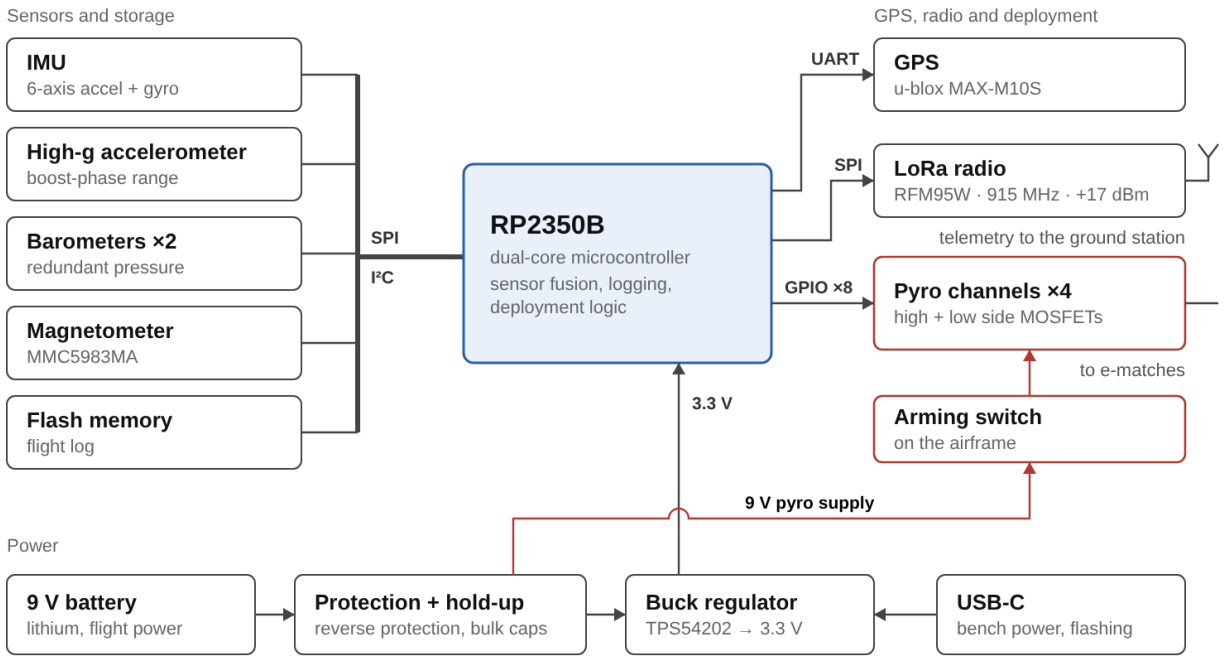
# DYLAN PIWKO

AEROSPACE ENGINEERING AT THE UNIVERSITY OF MICHIGAN

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## KESTREL FLIGHT COMPUTER

*Designer, Personal Project*
October 2026 – Present




*Kestrel block diagram*

### What?

- Flight computer for my 4" Wildman Punisher 4, which I'm flying for my NAR Level 2 certification in January 2027
- It logs data and sends live telemetry, and will eventually handle dual-deployment recovery, including a later L2200G flight that simulates to about 14,800 ft and Mach 1.7

### How?

- Designed the schematic and 4-layer board in **KiCad** (100 × 40 mm, 145 parts)
- RP2350B microcontroller, IMU, high-g accelerometer, two barometers, magnetometer, u-blox GPS, 915 MHz LoRa radio, and flash memory
- Four pyro channels switched on both the high and low side, behind a physical arming switch
- 9 V input with reverse protection and a TPS54202 buck converter for 3.3 V
- Wrote bring-up firmware and a desktop ground station with a live map

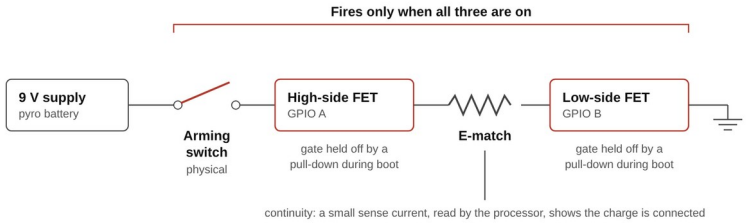
### Results

- Zero ERC/DRC errors, with every part checked pin by pin against its datasheet and Raspberry Pi's reference design
- Fixed everything an outside schematic review flagged
- PCBWay is building the boards through their Future Innovators sponsorship
- My commercial altimeters keep handling recovery until Kestrel passes ground and flight tests

## DESIGN AND VERIFICATION

I switched each e-match on both its high side and its low side, behind a physical arming switch on the airframe, so a charge can only fire when that switch is closed and the processor turns on both transistors.

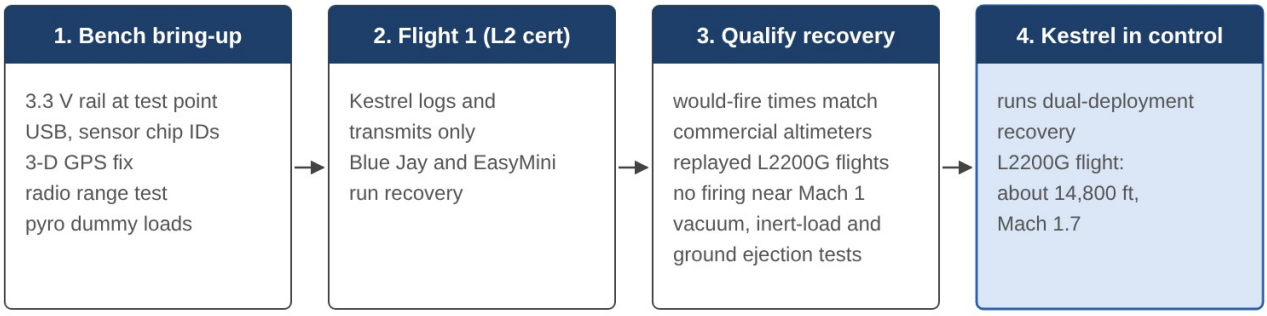
Pull-down resistors keep every gate off while the processor boots, and a small sense current through each e-match shows continuity without firing it.




*One of the four pyro channels*

I can only afford one board revision, so before ordering I checked each subsystem against a datasheet or reference design and planned a test to prove it once the boards show up.

| Subsystem | Checked against | Proven at bring-up by |
|---|---|---|
| Power | TI's TPS54202 3.3 V reference values; USB-only start-up margin | 3.25–3.35 V at the test point |
| Processor | All 80 RP2350B pins against the pin table and SparkFun's production board | Enumerates over USB |
| Sensors and GPS | Each pinout and SPI mode against its datasheet; u-blox integration manual | Chip ID self-test; 3-D GPS fix |
| Radio | Pinout; link budget of about 29 dB margin at 10 km | Ground-station range test |
| Pyro | MOSFET ratings, gate drive, boot pull-downs | Dummy-load tests, armed and disarmed |
| Layout | Zero ERC/DRC errors; outside review; PCBWay engineering review | Visual inspection of both assembled boards |

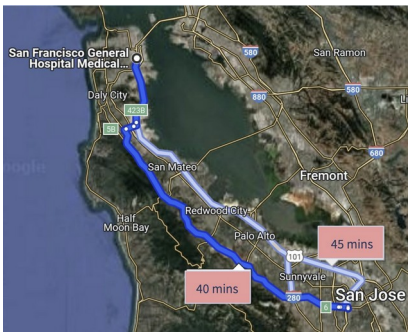



*Test plan before Kestrel controls recovery*

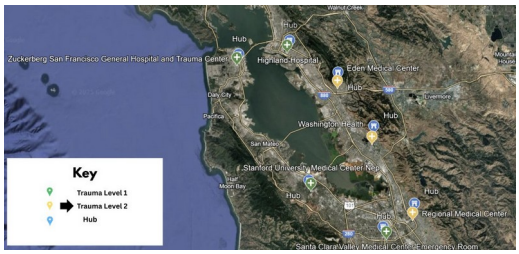
On the first flight Kestrel only logs and transmits. A Featherweight Blue Jay and an EasyMini handle recovery. Before I let it control a parachute, its would-fire times from flight 1 have to line up with when the commercial altimeters actually fired. It also has to fire correctly on replayed L2200G simulations without ever firing through the Mach transition. Then it goes through vacuum chamber, inert load, and ground ejection tests.

## AEROPULSE - NASA AMES RESEARCH CENTER

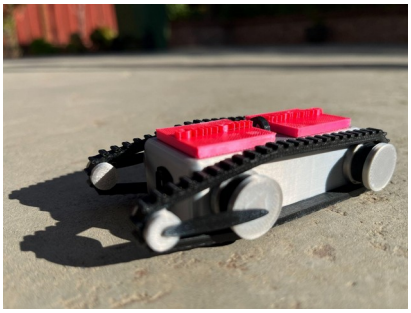
*Concept Design Intern*
June – August 2025




*Modeled route (map data © Google)*




*Proposed drone hubs (imagery © Google)*



*Sub-scale robot prototype*

### What?

- Our team of six designed **Aeropulse**, a concept that moves organs, lab samples, and urgent medicine between hospitals with hybrid VTOL drones and stair-climbing robots, with no manual handoffs
- I was one of 36 students picked from about 800 applicants for the Aviation Systems Division's Concept Design Experience

### How?

- I was responsible for drone routing and urban airspace integration
- Examined routing constraints, airspace rules (UTM, geofencing, detect-and-avoid), and trade-offs between the drones and the ground robots
- Conceptualized route planning for both vehicles, with 3-D occupancy-grid routing, GPS, and SLAM for the drones and A* and Dijkstra path planning for the robots
- Planned for operations below 400 ft AGL under **Part 107** with BVLOS waivers

### Results

- In our model, a donor heart gets from San Francisco General to Santa Clara Valley Medical Center in about 37 min, compared with 70 to 90+ min today
- Presented the design to 50+ NASA staff
- Co-authored NASA/TM-20250009029 with the rest of the team

### THE CONCEPT

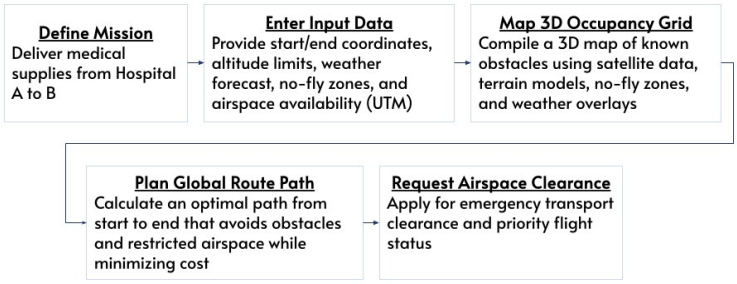



*A robot carries the payload out of Hospital A, rides the drone, and takes it to the right room in Hospital B (from the memo)*

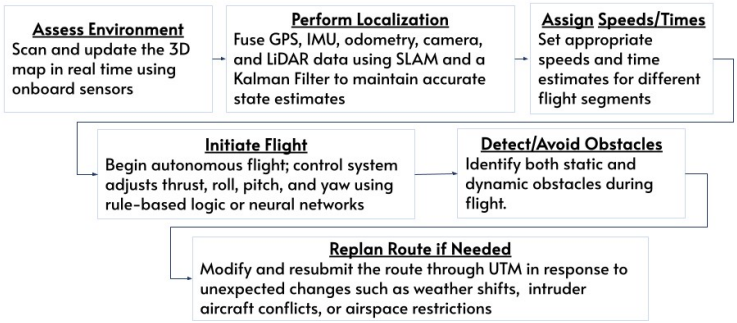
## ROUTE PLANNING (MY FOCUS AREA)

Before takeoff, the drone plans its whole route from Hospital A to Hospital B. It takes the start and end points, altitude limits, the weather forecast, no-fly zones, and open UTM airspace.

From those it builds a 3-D occupancy grid and finds the lowest-cost path around obstacles and restricted airspace. Then it asks UTM for emergency priority clearance.



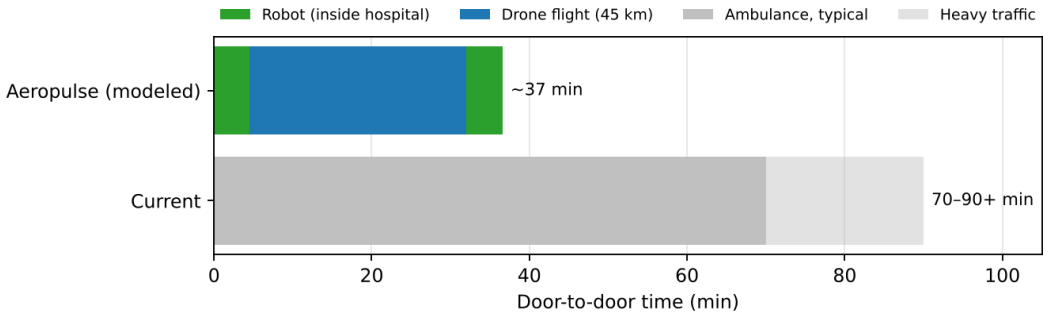

*Pre-flight route planning (from the memo)*




*In-flight route planning (from the memo)*

In the air, it keeps updating that map. It tracks its position by fusing GPS, IMU, odometry, camera, and LiDAR data with SLAM and a Kalman filter.

When something changes, like the weather or another aircraft getting in the way, it replans and sends the new route back through UTM. If it can't continue safely, it can loiter, land somewhere low-risk, return to base, or deploy a parachute.

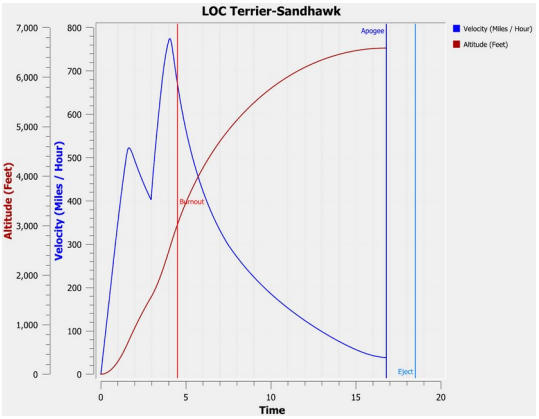



*Modeled heart transfer, San Francisco General to Santa Clara Valley Medical Center*

Full memo: [ntrs.nasa.gov/citations/20250009029 (NASA/TM-20250009029)](ntrs.nasa.gov/citations/20250009029)

# TERRIER-SANDHAWK TWO-STAGE ROCKET

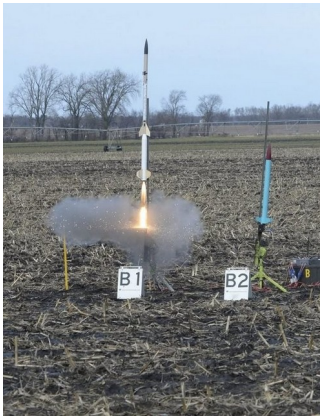
*Builder and Flyer, Personal Project*
January 2025




*RockSim simulation*




*Setting up on pad B1*




*Liftoff*

## What?

- My first two-stage high-power rocket, a LOC Precision 2.26" Sandhawk sustainer on a Terrier booster (76.9 in, 2,078 g)
- Flew it on a Cesaroni I297 and H180 at Michiana Rocketry on January 31, 2025

## How?

- Built the avionics for electronic staging, sustainer air-start, GPS tracking, and telemetry. The stages separate 1.0 s after booster burnout and the sustainer lights 0.7 s later
- Simulated the flight in **RockSim** to choose motors and check stability (2.40 static margin)
- Analyzed how the 0.7 s ignition delay, which RockSim doesn't model, changes the trajectory using a 2-D point-mass model

## Results

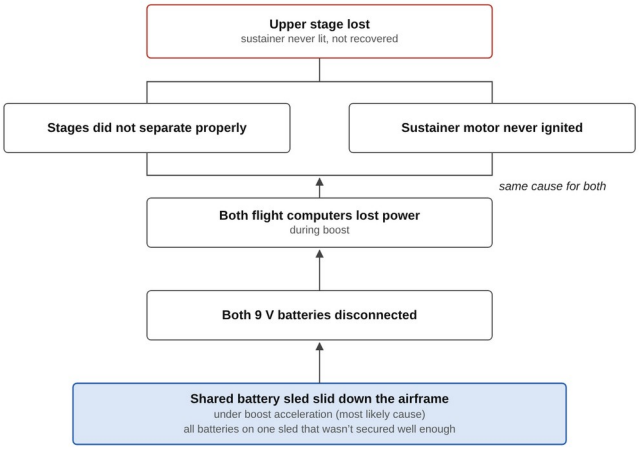
- The delay raised predicted apogee from about 6,590 ft to 6,860 ft
- On the flight the stages didn't separate properly, the sustainer never lit, and the upper stage was lost
- Both 9 V batteries were on one sled that most likely slid under boost and disconnected them, which taught me the importance of redundant avionics

## FAILURE ANALYSIS

The stages didn't separate properly and the sustainer never lit. Since both happened on the same flight, I looked for something that could cause both.

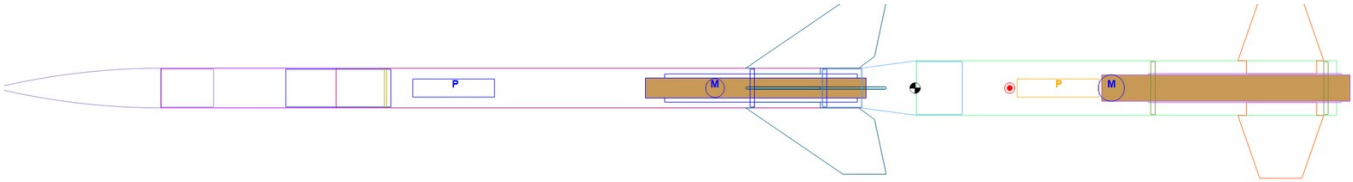
Every flight computer ran on a 9 V battery, and all of the batteries were mounted to the same sled. That sled wasn't secured well enough. Under boost acceleration it most likely slid down the airframe and disconnected both batteries, so all of the electronics lost power at once.

This flight taught me the importance of redundant avionics systems. Having more than one flight computer didn't help, because they all depended on the same mount.



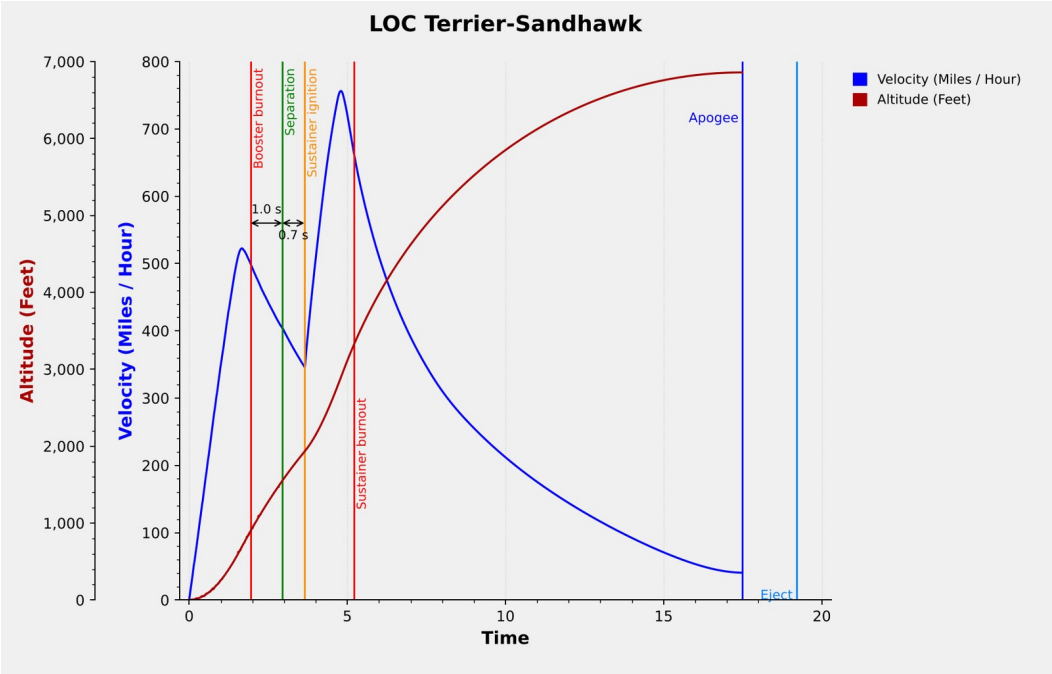

*How one loose sled led to both failures*

## STAGING DELAY ANALYSIS

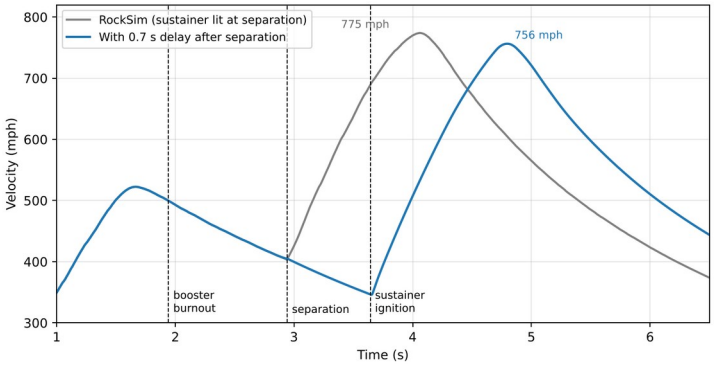



*RockSim model, sustainer on the left (CG 52.5 in, CP 57.9 in from the nose)*

RockSim lights the sustainer at the moment the stages separate, so its prediction does not include the 0.7 s wait before sustainer ignition. To model the actual sequence, the upper stage was re-flown from separation with a two-dimensional point-mass model. The sustainer's drag-to-mass ratio was taken from RockSim's coast phase as a function of Mach number, and its ignition mass was fitted to the boost phase using the H180's certified thrust curve. Run with RockSim's own timing, the model matches RockSim's apogee within 3 ft and its peak speed within 0.3%.




*Predicted flight with the 1.0 s separation and 0.7 s ignition delays*



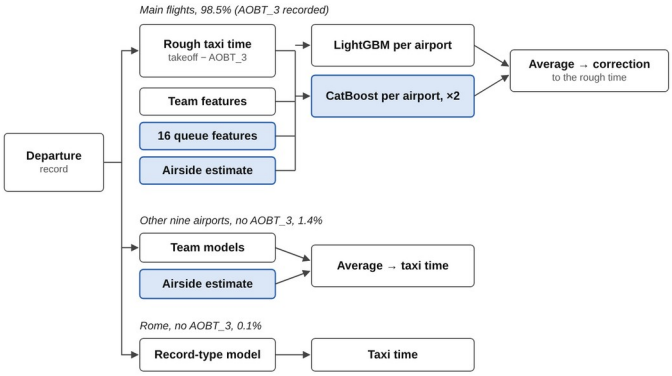

*RockSim (gray) vs. delayed ignition (blue)*

With the delay, the sustainer lights at 347 mph instead of 415 mph and peaks at 756 mph instead of 775 mph. Less of the burn happens near Mach 1, where drag is highest, so the predicted apogee rises from about 6,590 ft to 6,860 ft.

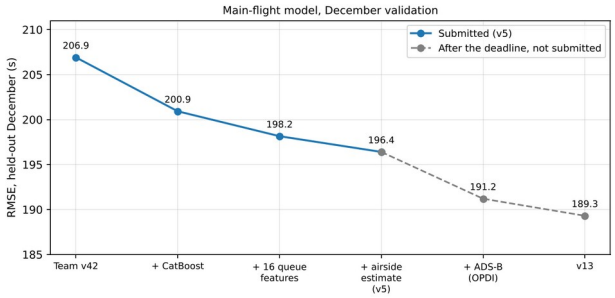
A 10% change in drag moves this by about 300 ft. Because the sustainer never lit on this flight, the prediction couldn't be checked against flight data.

# EUROCONTROL PRC DATA CHALLENGE

*Team Member, Embry-Riddle Team*
September – October 2026




*Team model, my additions in blue*




*Held-out error by version (gray points came after the deadline)*

## What?

- Predict aircraft taxi-out time at ten major European airports from about 2 million 2025 departures, scored by RMSE on January and July 2026 flights
- I joined Embry-Riddle's team "merry-llama" with very little Python or machine learning experience

## How?

- I was responsible for the model for main flights, which covers 98.5% of departures
- Added **CatBoost** models alongside the team's LightGBM models
- Built 16 departure-queue features, like how many planes took off from the same runway in the previous 5 to 20 minutes
- Used **SHAP** on the 1,000 worst predictions, then trained an "airside" taxi-time estimate that doesn't rely on the reported off-block time
- Decided what counted as an improvement before each test, and kept notes on what worked and what didn't across 20+ experiments

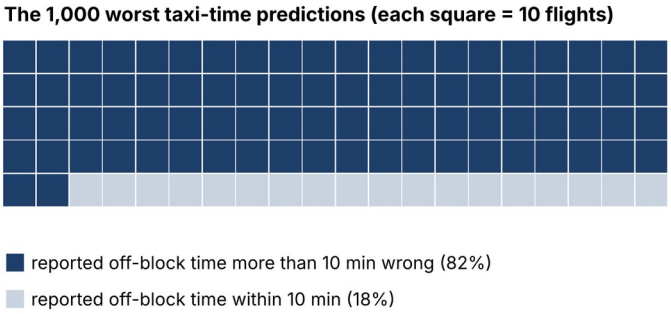
## Results

- Lowered the team's official RMSE from 276.53 s to 267.34 s
- Taeyun Yoo built the team's best score (265.02 s) and the final pipeline on top of this version. The final pipeline still uses my CatBoost models and airside estimate

## APPROACH

I started from the team's LightGBM model for main flights and added one thing at a time, testing each change on held-out data. Before each test I decided what result would count as an improvement.

When I looked at the worst predictions with SHAP, most of them were flights whose reported off-block time was more than 10 minutes off. So I trained an airside estimate that doesn't use that time at all.




*Worst 1,000 predictions*

Team code: [github.com/taeyunyoo/prc-2026-merry-llama](github.com/taeyunyoo/prc-2026-merry-llama)